WORKED SOLUTION

STEP II 2001, Question 3

Pure • Euclidean Geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q3

The key idea

Put the cuboid in coordinates with one vertex at the origin and the edges along the axes. Then every vertex has an obvious position vector, the diagonals are just differences of these, and the angle between them comes straight from the scalar product. The volume-to-surface-area ratio is a simple rational function of λ, and the final part links the two results by turning the condition on R into a condition on λ.

Solution

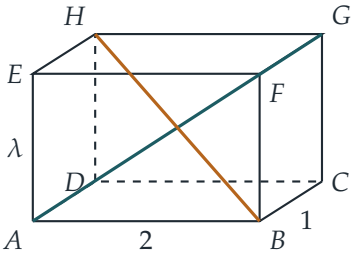
The angle between the diagonals

Since the edges AE, BF, CG, DH are perpendicular to the faces, all the angles at the vertices are right angles, so we may take axes along AB, AD and AE with A at the origin:

$$A = (0,0,0), \quad B = (2,0,0), \quad C = (2,1,0), \quad D = (0,1,0),$$

and the top face is the bottom face lifted by λ:

$$E = (0,0,\lambda), \quad F = (2,0,\lambda), \quad G = (2,1,\lambda), \quad H = (0,1,\lambda).$$



The two diagonals have direction vectors

$$\overrightarrow{AG} = \begin{pmatrix} 2 \\ 1 \\ \lambda \end{pmatrix}, \qquad \overrightarrow{BH} = \begin{pmatrix} -2 \\ 1 \\ \lambda \end{pmatrix},$$

each of length $\sqrt{5+\lambda^2}$, and

$$\overrightarrow{AG} \cdot \overrightarrow{BH} = -4 + 1 + \lambda^2 = \lambda^2 - 3.$$

If θ is the angle between these two vectors, then

$$\cos\theta = \frac{\lambda^2 - 3}{5 + \lambda^2}.$$

Ask yourself Where does the modulus sign come from?

Two lines that cross make two angles, θ and $\pi - \theta$, whose cosines are negatives of each other. The acute one (or the right angle, when $\lambda^2 = 3$) is the one with non-negative cosine, so

$$\cos\alpha = |\cos\theta| = \left|\frac{3 - \lambda^2}{5 + \lambda^2}\right|,$$

as required.

The ratio R

The volume is $2 \times 1 \times \lambda = 2\lambda$. The surface area consists of two faces of each of the areas 2×1, $2 \times \lambda$ and $1 \times \lambda$, so it is $2(2 + 3\lambda)$. Hence

$$R = \frac{2\lambda}{2(2+3\lambda)} = \frac{\lambda}{2+3\lambda}.$$

Since $\lambda > 0$, the denominator is positive, and

$$R < \tfrac{1}{3} \iff 3\lambda < 2 + 3\lambda \iff 0 < 2,$$

which is true. So $R < \frac{1}{3}$ for every possible λ. (As $\lambda \to \infty$, $R \to \frac{1}{3}$, so the bound cannot be improved.)

The final part

Again because $2 + 3\lambda > 0$, we may multiply through:

$$R \geq \tfrac{1}{4} \iff 4\lambda \geq 2 + 3\lambda \iff \lambda \geq 2.$$

So suppose $\lambda \geq 2$. Then $\lambda^2 \geq 4 > 3$, so $\lambda^2 - 3 > 0$ and the modulus can be dropped:

$$\cos\alpha = \frac{\lambda^2 - 3}{\lambda^2 + 5} = 1 - \frac{8}{\lambda^2 + 5}.$$

Writing it this way shows that $\cos\alpha$ increases as λ^2 increases. With $\lambda^2 \geq 4$ we have $\lambda^2 + 5 \geq 9$, so

$$\cos\alpha \geq 1 - \frac{8}{9} = \frac{1}{9}.$$

Both α and $\arccos\frac{1}{9}$ lie in $[0, \frac{\pi}{2}]$, where cosine is strictly decreasing, so $\cos\alpha \geq \frac{1}{9}$ gives

$$\alpha \leq \arccos\tfrac{1}{9},$$

which is what we had to prove. Equality holds exactly when $\lambda = 2$, i.e. when $R = \frac{1}{4}$.

Checks

- When $\lambda^2 = 3$ the formula gives $\cos\alpha = 0$. Check: $\overrightarrow{AG} \cdot \overrightarrow{BH} = -4 + 1 + 3 = 0$, so the diagonals really are perpendicular.
- As $\lambda \to \infty$ the cuboid becomes a tall thin prism, and both diagonals become nearly vertical, so $\alpha \to 0$; the formula gives $\cos\alpha \to 1$, as it should.
- At $\lambda = 2$: volume 4, surface area $2(2 + 4 + 2) = 16$, so $R = \frac{1}{4}$, and $\cos\alpha = \frac{1}{9}$, so $\alpha \approx 83.6°$. A numerical check over 10^5 random values of λ confirmed the angle formula, $R < \frac{1}{3}$, and $\alpha \leq \arccos\frac{1}{9}$ whenever $R \geq \frac{1}{4}$.

Common slips

- Forgetting the modulus: for $\lambda^2 < 3$ the vectors $\overrightarrow{AG}$ and $\overrightarrow{BH}$ make an obtuse angle, and the acute angle is its supplement.
- Getting the direction of the last inequality wrong: a *larger* cosine means a *smaller* angle, because cosine decreases on $[0, \frac{\pi}{2}]$.

Beyond STEP

Placing an object in coordinates and using scalar products to find angles is the basic tool of vector geometry, and in a first-year linear algebra course it generalises to inner products in any number of dimensions: the angle between two vectors in $\mathbb{R}^n$ is *defined* by $\cos\theta = \mathbf{u} \cdot \mathbf{v} / (|\mathbf{u}||\mathbf{v}|)$, with the Cauchy–Schwarz inequality guaranteeing that this lies in $[-1, 1]$. A nice exercise is to show that a long diagonal of an n-dimensional unit cube makes an angle with each edge whose cosine is $1/\sqrt{n}$, so the angle tends to a right angle as n grows. Strang's *Introduction to Linear Algebra* develops these ideas clearly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.